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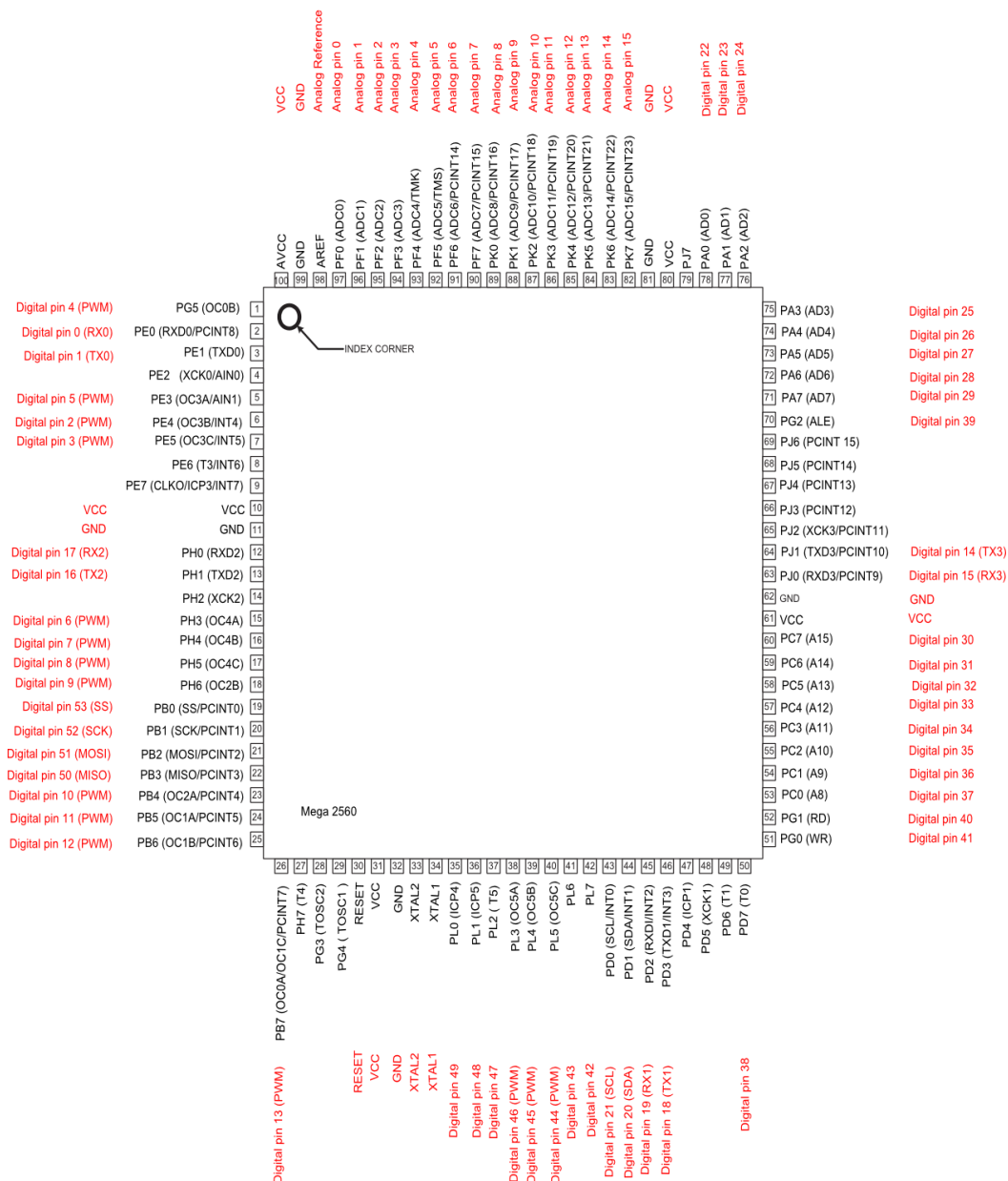


Abbildung 1 ATMEGA2560 PINOUT

Load bootloader with Arduino Uno

Connect Arduino Uno with ATMEGA 2560

Arduino UNO SS (D10) -> ATMEGA2560 RESET Pin (30)

Arduino UNO MOSI (D11) -> ATMEGA2560 MOSI (21)

Arduino UNO MISO (D12) -> ATMEGA2560 MISO (22)

Arduino UNO SCK (D13) -> ATMEGA2560 SCK (20)

Arduino UNO VCC 5V -> ATMEGA2560 5V (31)

Arduino UNO GND -> ATMEGA2560 GND (32)

16.000 MHz Crystal left leg XTAL2 (33) right leg XTAL1 (34)

3x 0.1uf capacitor close to the chip between VCC and GND

[Open bootloader](#) code (from nickgammon) in the Arduino IDE

1. Load code on Arduino Uno

2. Open Serialmonitor

```
Atmega chip programmer.
Written by Nick Gammon.
Entered programming mode OK.
Signature = 0x1E 0x98 0x01
Processor = ATmega2560
Flash memory size = 262144 bytes.
LFuse = 0xFF
HFuse = 0xD8
EFuse = 0xFD
Lock byte = 0xCF
Bootloader address = 0x3E000
Bootloader length = 8192 bytes.
Type 'G' to program the chip with the bootloader ..
```

3. If this appears everything is correct

4. Press G and Enter

```
Erasing chip ...
Writing bootloader ...
Committing page starting at 0x3E000
Committing page starting at 0x3E100
Committing page starting at 0x3E200
Committing page starting at 0x3E300
Committing page starting at 0x3E400
Committing page starting at 0x3E500
[...]
Committing page starting at 0x3E600
Committing page starting at 0x3FA00
Committing page starting at 0x3FB00
Committing page starting at 0x3FC00
Committing page starting at 0x3FD00
Committing page starting at 0x3FE00
Committing page starting at 0x3FF00
Written.
Verifying ...
No errors found.
Writing fuses ...
LFuse = 0xFF
HFuse = 0xD8
EFuse = 0xFD
Lock byte = 0xCF
Done.
Type 'C' when ready to continue with another chip ...
```

5. If this appears everything is correct and the bootloader is loaded

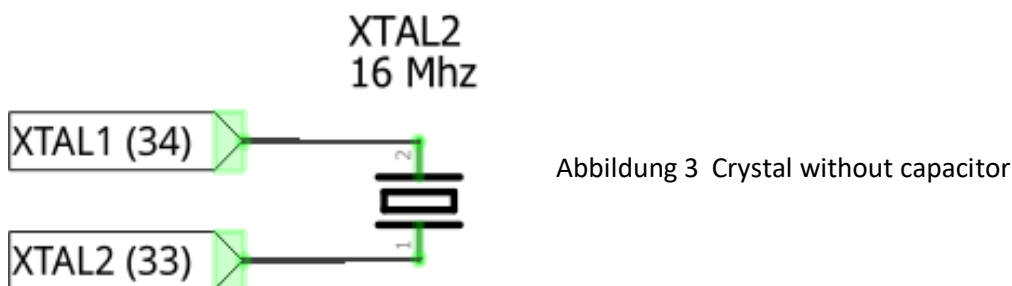
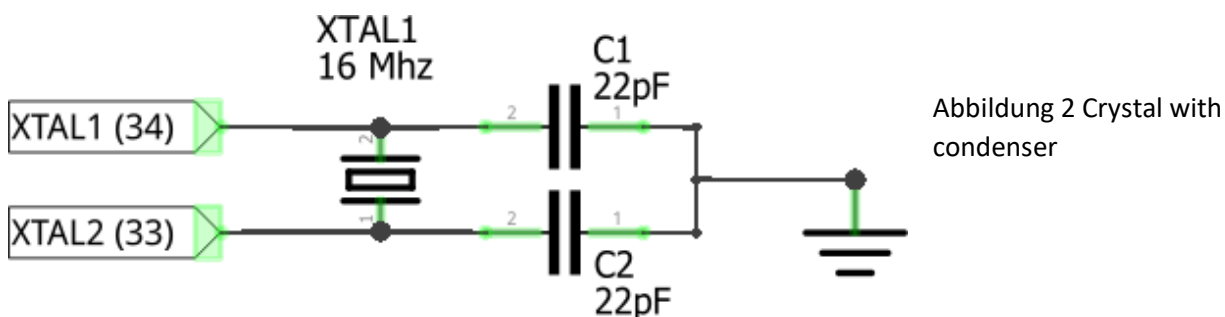
6. Remove wiring to Arduino Uno

Load code with CP2102

Connect CP2102 -> ATMEGA2560
0.1uF between CP2101 DTR pin and RESET (pin 30)
10K resistor to 5V and RESET (Pin30)
CP2101 RX -> ATMEGA2560 TX Pin (3)
CP2101 TX -> ATMEGA2560 RX Pin (2)
CP2101 5V -> ATMEGA2560 VCC Pin (31+100)
CP2101 GND -> ATMEGA2560 GND Pin (32+99)
22pf capacitor between ATMEGA2560 XTAL2 (33) and GND
22pf capacitor between ATMEGA2560 XTAL1 (34) and GND
16.000 MHz Crystal left leg XTAL2 (33) right leg XTAL1 (34)
Install CP2102 driver
1. 7. write code in Arduino IDE 2. 8. connect CP2102 with PC 3. 9. specify in Arduino IDE as Arduino Mega2560 4. 10. upload code

Wiring for general operation

VCC 5V to pin 10, 31, 61, 80, 100
GND to pin 11, 32, 62, 81, 99
22pf capacitor between XTAL2 (33) and GND
22pf capacitor between XTAL1 (34) and GND
16.000 MHz Crystal left leg XTAL2 (33) right leg XTAL1 (34)
10K resistor at 5V and RESET (Pin30)
3x 0.1uf capacitor close to the chip between VCC and GND



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Sources

<http://www.gammon.com.au/bootloader>

https://github.com/nickgammon/arduino_sketches

<https://kevinfundarek.wordpress.com/2015/11/10/programming-the-atmega-2560-with-the-arduino-ide/>

<https://www.instructables.com/DIY-Bare-Minimum-Arduino-Mega-2560/>

Image sources

<https://forum.arduino.cc/index.php?topic=615812.0>

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